HF Scaling

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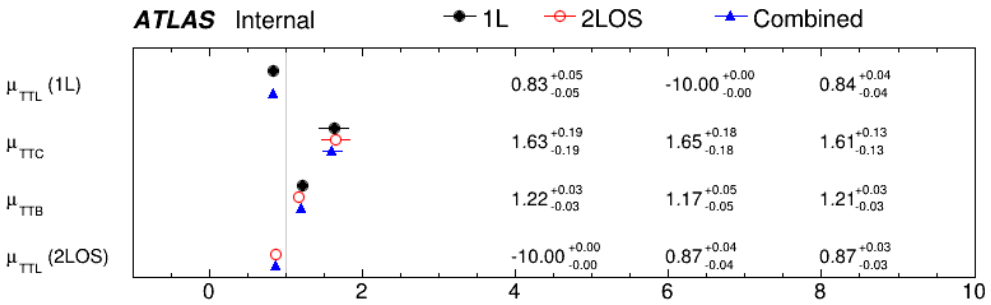
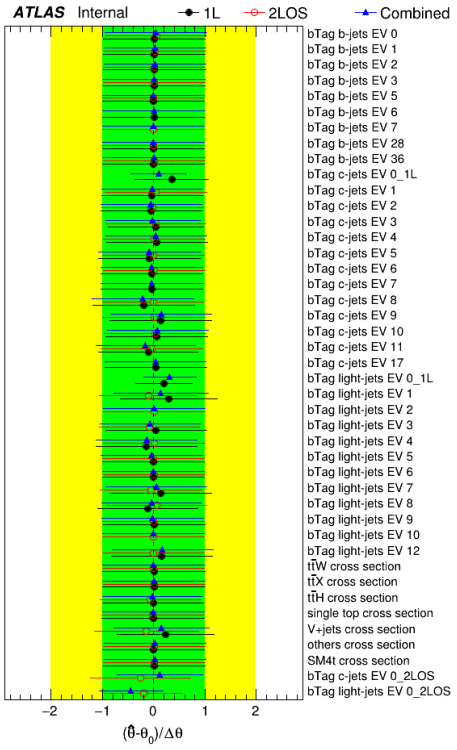
Introduction

- Getting the HF-correction factors $\mu_{TTL}(1L, 2LOS)$, μ_{TTC} and μ_{TTB})
- Combine fit with 1L and 2LOS channel
- Decorrelating the C0 and L0 with different channels
- Removing TTL XS systematics

Nominal Sample





Nominal Sample

PreFit 1L

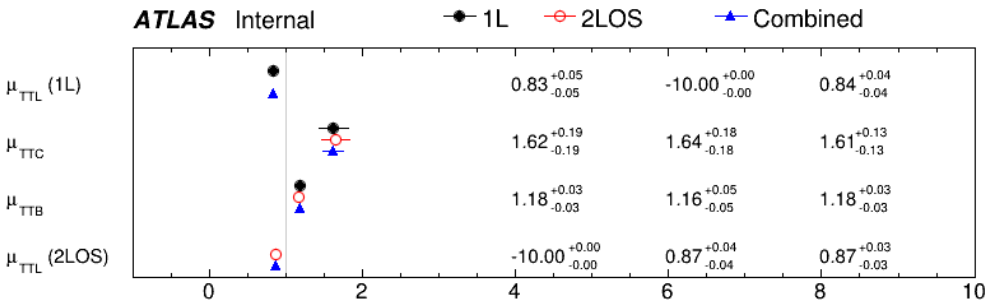
	$LJETS, 7j, 2b$	$LJETS, 7j, 3b$	$LJETS, 7j, \geq 4b$	Total
TTL	132532.64 ± 4215.99	5751.48 ± 470.26	44.24 ± 11.26	138328.36 ± 4242.15
TTC	32634.67 ± 888.81	4491.35 ± 360.70	181.26 ± 27.45	37307.28 ± 959.60
TTB	14525.10 ± 17.06	9577.63 ± 12.25	2016.50 ± 4.30	26119.23 ± 21.44
PostFit 1L				
TTL	110383.95 ± 4098.93	4719.78 ± 185.43	34.49 ± 2.65	115138.22 ± 4103.12
TTC	52103.97 ± 4091.81	7103.34 ± 553.42	284.09 ± 22.62	59491.40 ± 4129.13
TTB	17506.67 ± 393.54	11543.64 ± 259.49	2430.43 ± 54.63	31480.74 ± 474.55

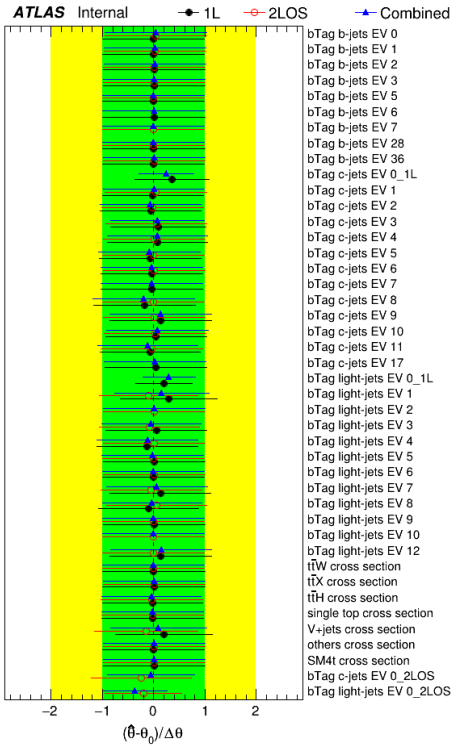
PreFit 2LOS

	$OS2L, 5j, 2b$	$OS2L, 5j, 3b$	$OS2L, 5j, \geq 4b$	Total
TTL	31869.24 ± 961.92	189.42 ± 38.91	0.51 ± 0.26	32059.17 ± 962.71
TTC	6015.13 ± 150.02	645.69 ± 52.96	10.95 ± 1.71	6671.77 ± 159.10
TTB	2994.50 ± 5.91	1870.36 ± 4.24	296.20 ± 1.36	5161.06 ± 7.40
PostFit 2LOS				
TTL	27827.39 ± 790.84	179.33 ± 26.33	0.53 ± 0.18	28007.25 ± 791.28
TTC	9680.28 ± 701.43	1031.59 ± 63.60	17.64 ± 1.60	10729.51 ± 704.31
TTB	3609.30 ± 81.15	2254.41 ± 50.68	357.04 ± 8.05	6220.75 ± 96.01

1L Post/Pre Ratio: TTL=0.83 ; TTC=1.59 ; TTB=1.21
2LOS Post/Pre Ratio: TTL=0.87 ; TTC=1.61 ; TTB=1.21
Combined Post/Pre Ratio: TTL=0.84 ; TTC=1.60 ; TTB=1.21

ttbb 4FS Sample





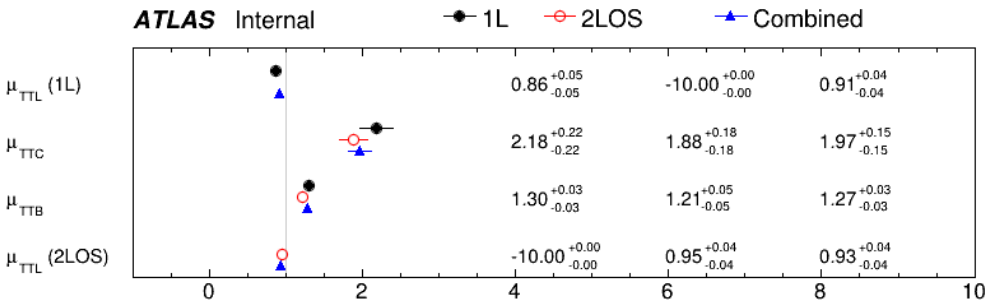
ttbb 4FS Sample

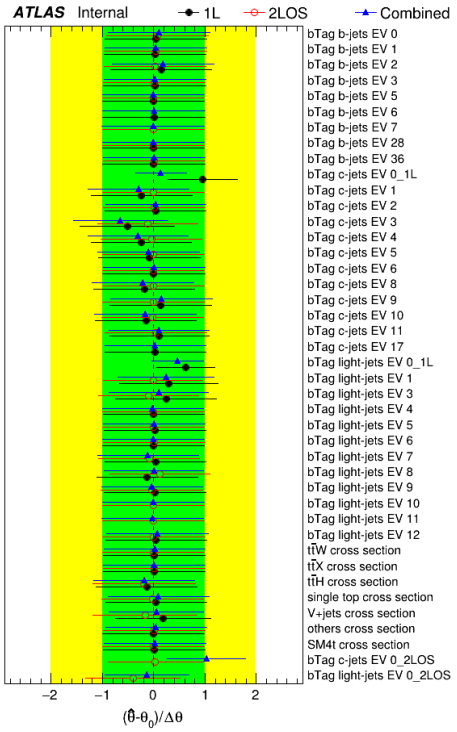
PreFit 1L (4FS)				
	$LJETS, 7j, 2b$	$LJETS, 7j, 3b$	$LJETS, 7j, \geq 4b$	Total
TTL	132532.64 ± 4215.99	5751.48 ± 470.26	44.24 ± 11.26	138328.36 ± 4242.15
TTC	32634.67 ± 888.81	4491.35 ± 360.70	181.26 ± 27.45	37307.28 ± 959.60
TTB	14949.35 ± 21.55	9905.66 ± 17.62	2078.26 ± 8.84	26933.27 ± 29.21
PostFit 1L (Nominal)				
TTL	110383.95 ± 4098.93	4719.78 ± 185.43	34.49 ± 2.65	115138.22 ± 4103.12
TTC	52103.97 ± 4091.81	7103.34 ± 553.42	284.09 ± 22.62	59491.40 ± 4129.13
TTB	17506.67 ± 393.54	11543.64 ± 259.49	2430.43 ± 54.63	31480.74 ± 474.55

PreFit 2LOS (4FS)				
	$OS2L, 5j, 2b$	$OS2L, 5j, 3b$	$OS2L, 5j, \geq 4b$	Total
TTL	31869.24 ± 961.92	189.42 ± 38.91	0.51 ± 0.26	32059.17 ± 962.71
TTC	6015.13 ± 150.02	645.69 ± 52.96	10.95 ± 1.71	6671.77 ± 159.10
TTB	3052.65 ± 7.23	1893.49 ± 5.59	297.81 ± 2.44	5243.95 ± 9.46
PostFit 2LOS (Nominal)				
TTL	27827.39 ± 790.84	179.33 ± 26.33	0.53 ± 0.18	28007.25 ± 791.28
TTC	9680.28 ± 701.43	1031.59 ± 63.60	17.64 ± 1.60	10729.51 ± 704.31
TTB	3609.30 ± 81.15	2254.41 ± 50.68	357.04 ± 8.05	6220.75 ± 96.01

1L Post/Pre Ratio: TTL=0.83 ; TTC=1.59 ; TTB=1.21
2LOS Post/Pre Ratio: TTL=0.87 ; TTC=1.61 ; TTB=1.21
Combined Post/Pre Ratio: TTL=0.84 ; TTC=1.60 ; TTB=1.17

aMcAtNloPy8 Sample





aMcAtNloPy8 Sample

PreFit 1L (aMcAtNloPy8)

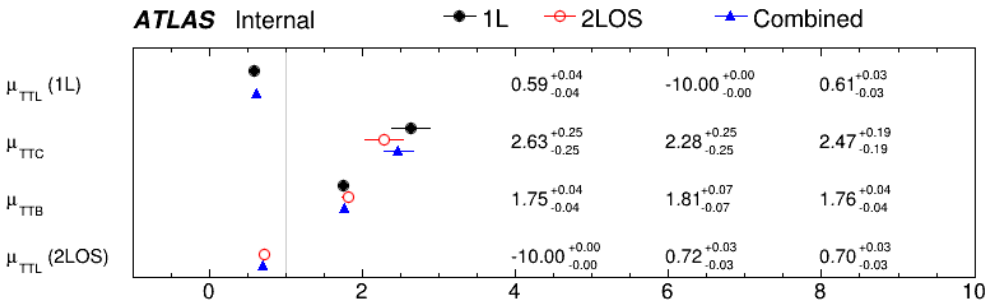
	$LJETS, 7j, 2b$	$LJETS, 7j, 3b$	$LJETS, 7j, \geq 4b$	Total
TTL	117273.46 ± 3716.78	5116.16 ± 420.24	38.65 ± 10.28	122428.27 ± 3740.48
TTC	29348.28 ± 809.43	3924.70 ± 314.28	151.22 ± 22.69	33424.20 ± 868.60
TTB	13398.83 ± 38.45	8851.56 ± 27.21	1913.43 ± 8.88	24163.82 ± 47.93
PostFit 1L (Nominal)				
TTL	110383.95 ± 4098.93	4719.78 ± 185.43	34.49 ± 2.65	115138.22 ± 4103.12
TTC	52103.97 ± 4091.81	7103.34 ± 553.42	284.09 ± 22.62	59491.40 ± 4129.13
TTB	17506.67 ± 393.54	11543.64 ± 259.49	2430.43 ± 54.63	31480.74 ± 474.55

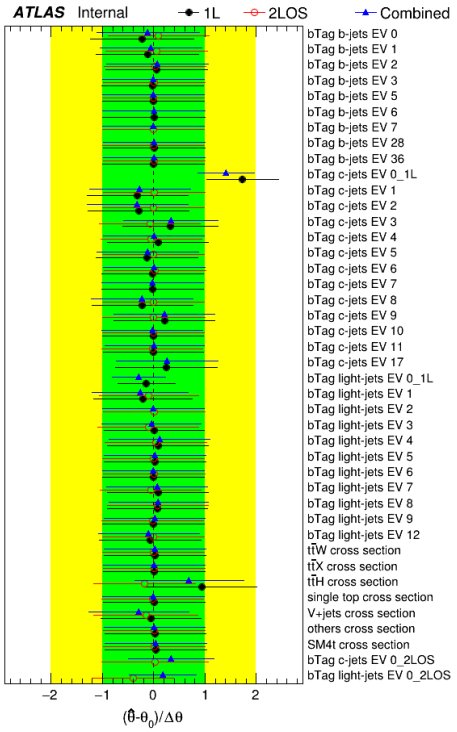
PreFit 2LOS (aMcAtNloPy8)

	$OS2L, 5j, 2b$	$OS2L, 5j, 3b$	$OS2L, 5j, \geq 4b$	Total
TTL	28880.01 ± 889.56	160.68 ± 33.08	0.72 ± 0.40	29041.41 ± 890.17
TTC	5565.00 ± 140.81	593.38 ± 48.86	9.58 ± 1.60	6167.96 ± 149.05
TTB	2828.95 ± 11.79	1772.51 ± 8.41	296.82 ± 2.72	4898.28 ± 14.74
PostFit 2LOS (Nominal)				
TTL	27827.39 ± 790.84	179.33 ± 26.33	0.53 ± 0.18	28007.25 ± 791.28
TTC	9680.28 ± 701.43	1031.59 ± 63.60	17.64 ± 1.60	10729.51 ± 704.31
TTB	3609.30 ± 81.15	2254.41 ± 50.68	357.04 ± 8.05	6220.75 ± 96.01

1L Post/Pre Ratio: TTL=0.94 ; TTC=1.78 ; TTB=1.30
2LOS Post/Pre Ratio: TTL=0.96 ; TTC=1.74 ; TTB=1.27
Combined Post/Pre Ratio: TTL=0.95 ; TTC=1.77 ; TTB=1.30

PhHerwig Sample





PhHerwig Sample

PreFit 1L (PhHerwig)

	$LJETS, 7j, 2b$	$LJETS, 7j, 3b$	$LJETS, 7j, \geq 4b$	Total
TTL	166159.84 ± 5232.84	7001.95 ± 573.57	52.10 ± 13.60	173213.89 ± 5264.20
TTC	23306.84 ± 635.75	3190.58 ± 255.02	125.72 ± 18.59	26623.14 ± 685.24
TTB	11965.54 ± 19.36	7104.79 ± 13.27	1346.67 ± 4.30	20417.00 ± 23.86
PostFit 1L (Nominal)				
TTL	110383.95 ± 4098.93	4719.78 ± 185.43	34.49 ± 2.65	115138.22 ± 4103.12
TTC	52103.97 ± 4091.81	7103.34 ± 553.42	284.09 ± 22.62	59491.40 ± 4129.13
TTB	17506.67 ± 393.54	11543.64 ± 259.49	2430.43 ± 54.63	31480.74 ± 474.55

PreFit 2LOS (PhHerwig)

	$OS2L, 5j, 2b$	$OS2L, 5j, 3b$	$OS2L, 5j, \geq 4b$	Total
TTL	38019.17 ± 1160.60	199.01 ± 40.78	0.95 ± 0.46	38219.13 ± 1161.32
TTC	4689.39 ± 118.92	494.01 ± 40.09	7.17 ± 1.09	5190.57 ± 125.50
TTB	1845.12 ± 5.22	1184.89 ± 3.62	204.56 ± 1.20	3234.57 ± 6.46
PostFit 2LOS (Nominal)				
TTL	27827.39 ± 790.84	179.33 ± 26.33	0.53 ± 0.18	28007.25 ± 791.28
TTC	9680.28 ± 701.43	1031.59 ± 63.60	17.64 ± 1.60	10729.51 ± 704.31
TTB	3609.30 ± 81.15	2254.41 ± 50.68	357.04 ± 8.05	6220.75 ± 96.01

1L Post/Pre Ratio: TTL=0.67 ; TTC=2.23 ; TTB=1.54
2LOS Post/Pre Ratio: TTL=0.73 ; TTC=2.07 ; TTB=1.92
Combined Post/Pre Ratio: TTL=0.68 ; TTC=2.21 ; TTB=1.59